

Majorana Modes in the Machine

Week 9: Circuit-Level Gate Noise on the VQE – Depth Scaling and Verification

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From Week 7 to Week 9

Week 7 gave us

- A validated parity-constrained VQE: optimal $\theta^*(\mu)$ across the phase diagram.
- A depth diagnostic: *noiselessly*, subspace infidelity *falls* as the ansatz repetitions r grow.
- Conclusion then: deeper ansatz = better state.

Same Hamiltonian and edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ as Block 3; the new ingredient is **noise that enters through the gates**.

Week 9 asks

- Put *real gate noise* into that preparation circuit.
- Each repetition adds an entangling layer; every CNOT carries an error channel.
- How does the edge-string diagnostic survive as the circuit deepens?

Gate Errors as Quantum Channels

A noisy gate is the ideal unitary followed by a CPTP error channel:

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \quad \mathcal{E}_g(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I. \quad (1)$$

A full circuit composes these gate-by-gate:

$$\rho_{\text{out}} = (\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}) \circ \cdots \circ (\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1})(\rho_{\text{in}}). \quad (2)$$

Key consequence

Noise enters *through the gates*: the number of error channels grows with the gate count, hence with the circuit depth. A density-matrix simulation with a gate-level `NoiseModel` captures this exactly.

Noise Integrated Through the Gates, by reps

The linear EfficientSU2 ansatz at depth r has

$$\# \text{ CNOTs} = r(L - 1),$$

and each one applies a depolarizing channel. The *measured* signal is a product of two opposing factors:

$$|\langle O_{\text{edge}} \rangle|_{\text{meas}}(r) \approx \underbrace{|\langle O_{\text{edge}} \rangle|_{\text{ideal}}(r)}_{\text{expressibility } \uparrow \text{ (Week 7)}} \times \underbrace{(1 - p_{\text{cx}})^{r(L-1)}}_{\text{gate noise } \downarrow}. \quad (3)$$

- Week 7: the first factor *rises* toward 1 with depth.
- Week 9: the second factor *falls* exponentially in depth.

The depth trade-off

Deeper circuits prepare a better state but accumulate more gate noise. There is an **optimal ansatz depth** on a noisy device.

Integrating Gate Noise in Qiskit

```
from qiskit_aer import AerSimulator
from qiskit_aer.noise import NoiseModel, depolarizing_error

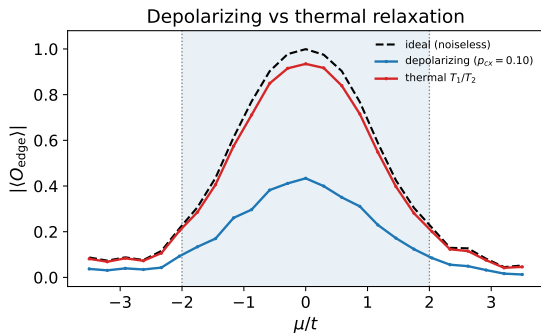
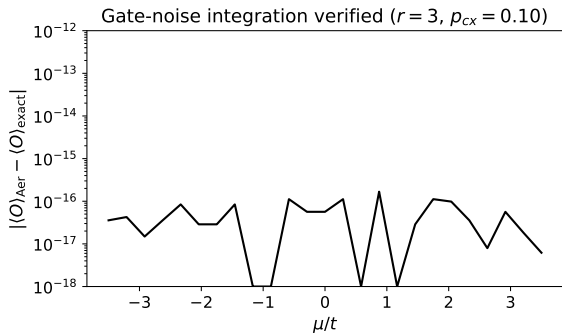
nm = NoiseModel()
nm.add_all_qubit_quantum_error(depolarizing_error(p_cx, 2), ['cx'])

sim = AerSimulator(method='density_matrix', noise_model=nm)
qc = ansatz.assign_parameters(theta)    # the Week 7 VQE state prep
qc.save_density_matrix()
rho = sim.run(transpile(qc, sim)).result().data()['density_matrix']
edge = rho.expectation_value(0_edge).real
```

- `add_all_qubit_quantum_error` fires the channel after every `cx` in the transpiled circuit.
- Reading $\text{Tr}[O_{\text{edge}}\rho]$ from the density matrix avoids shot noise while we study the gate layer.

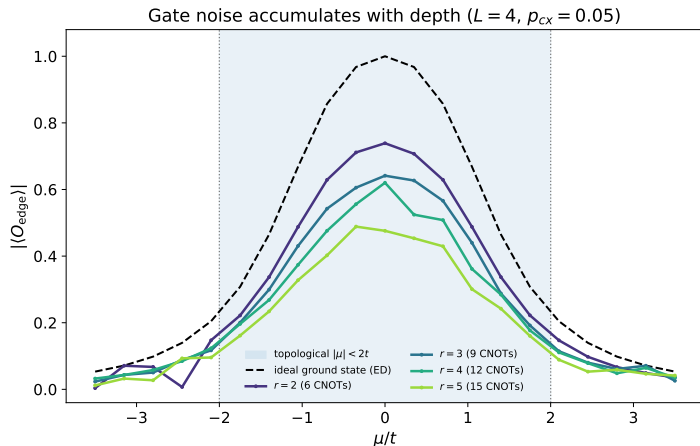
Precise Verification + a Physical T_1/T_2 Model

Verified circuit-level gate noise and a physical T_1/T_2 model



Left: the Aer NoiseModel edge string vs. an *independent* exact gate-by-gate density-matrix evolution (same circuit, depolarizing SuperOp after each cx) agree to $\sim 10^{-15}$ across all μ – the gate noise is correctly integrated through the whole circuit. **Right:** swapping depolarizing for `thermal_relaxation_error( $T_1, T_2, \tau$ )` gives the realistic channel; T_1 additionally poisons the parity sector, T_2 kills the endpoint coherence the edge string needs.

Gate Noise vs. Depth Across the Phase Diagram



Same per-cx strength at every depth; only the CNOT count $r(L - 1)$ changes.

- The noiseless curve (dashed) marks the topological plateau $|\mu| < 2t$.
- As r grows, the measured edge-string sweep is pushed *down*: the diagnostic contrast flattens.
- The phase distinction is most robust at the $\mu = 0$ sweet spot and shallow depth.

Summary and What This Enables Next

- Gate noise is now integrated at the *circuit* level via a Qiskit `NoiseModel`, and verified to machine precision against an exact gate-by-gate reference.
- The edge-string diagnostic degrades with ansatz depth: the cost of expressibility is accumulated CNOT noise, giving an optimal working depth.
- A physical T_1/T_2 relaxation model is in place as the realistic upgrade to depolarizing.

Next

- **Noisy VQE optimization:** optimize with a `NoiseModel`-backed estimator, so every cost evaluation is corrupted.
- **Then** return to the professor's questions: does topology protect the parity, and how does chain length interact with all of this.

Code: `src/block4.py` (plots 2–3). Notes: `notes/week9_note.tex`.